

THE MANY FLAVOURS OF REGRESSION

June 2025



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The Many Flavours of Regression

GENERALIZED ADDITIVE MODELS (GAMS)

Non-linear relationships

What is meant by non-linear?

- Non-linear in the parameters

Use a smooth function to link response and predictor

$$y_i = \beta_0 + f(x_i) + \epsilon_i$$



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CASE STUDY

What affects the body condition of fin whales in the North Atlantic?

- Data collected from 1967-2010
- Full analysis in Williams et al. 2013

Variables:

- **LE** - length in meters
- **m4** - a measure of body condition
- **NewStatus** - Pregnant (1) or not (0)
- **SelP** - Zooplankton from Selvogsbanki



SPLINES

Definitions:

- A thin wood or metal strip used in building construction
- A numeric function that is piecewise-defined by polynomial functions



HOW TO CONNECT THE SPLINES?

Knots!

- Number of knots important
- Defines flexibility of the line
- Penalized smoothing spline
 - Cost to being non-linear
 - Avoid over-fitting



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The Many Flavours of Regression

IN R

```
1 library(mgcv)
2 fin <- gam(m4~s(LE,bs="ts")+s(SelP,bs="ts")+NewStatus,data=Fin)
3 summary(fin)
```

Family: gaussian
Link function: identity

Formula:
m4 ~ s(LE, bs = "ts") + s(SelP, bs = "ts") + NewStatus

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	58.627	1.052	55.754	< 2e-16 ***
NewStatus	9.618	1.308	7.352	7.38e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
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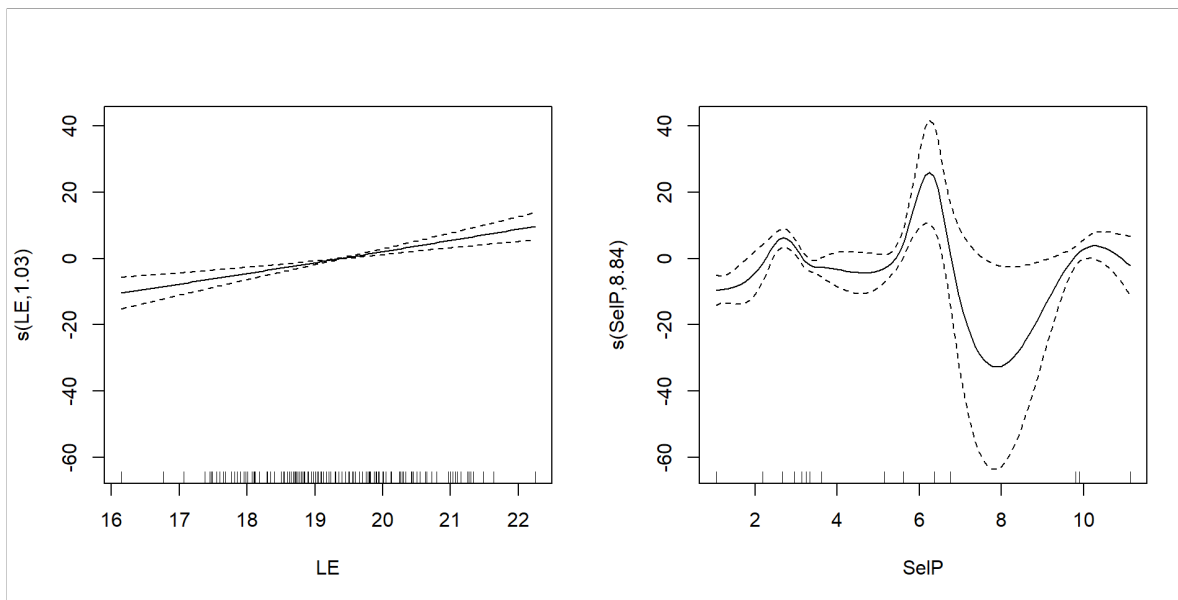


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VISUAL

```
1 par(mfrow=c(1,2))
2 plot(fin)
```



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DIAGNOSE

```
1 par(mfrow=c(2,2))
2 gam.check(fin)
```

```
Method: GCV   Optimizer: magic
Smothing parameter selection converged after 12 iterations.
The RMS GCV score gradient at convergence was 0.001029153 .
The Hessian was positive definite.
Model rank = 20 / 20

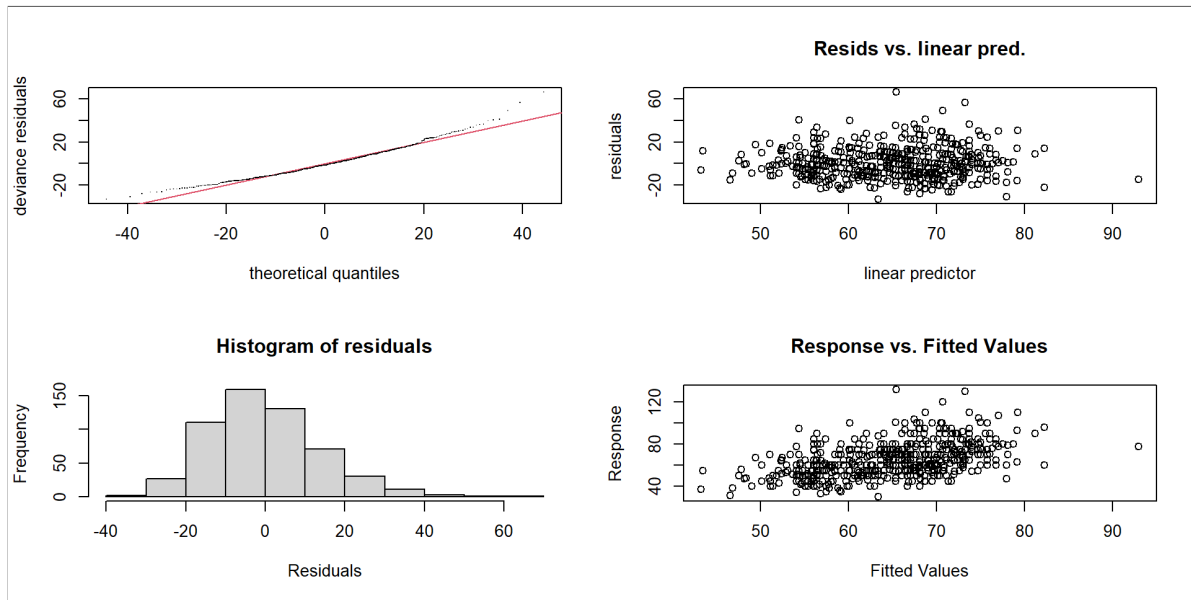
Basis dimension (k) checking results. Low p-value (k-index<1) may
indicate that k is too low, especially if edf is close to k'.

      k'  edf k-index p-value
s(LE)  9.00 1.03  1.07  0.97
s(SelP) 9.00 8.84  0.95  0.09 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

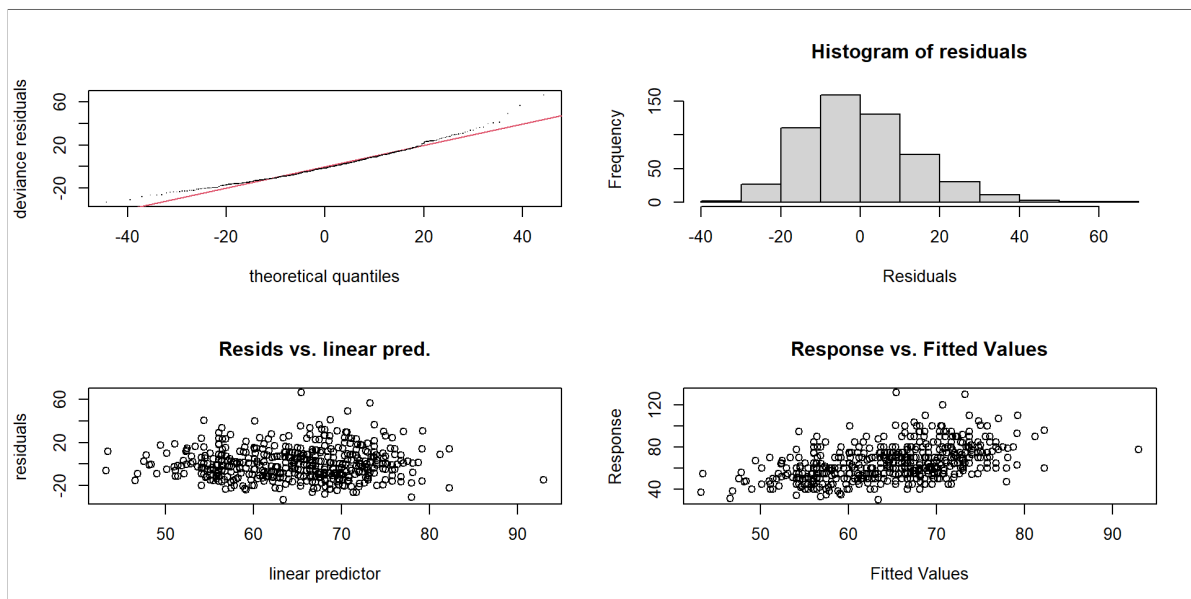


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DIAGNOSTIC PLOTS



MORE ON GAMS

Change the family Add random effects (GAMM)

Great online tutorial at: <https://noamross.github.io/gams-in-r-course/>



GENERALIZED ESTIMATING EQUATIONS (GEES)

Method for modelling longitudinal or clustered data

Why not a mixed model?

- GEE models population average, MM are subject specific models
- GEE doesn't use a likelihood so can be easier to compute
- Can specify the correlation structure



WHY NOT A GEE?

- Doesn't handle complex study designs well
- Model not suitable for simulation
- A lot of user definition required

geepack in R for implementation



RIDGE REGRESSION

Addresses two issues:

- Number of parameters exceeds the number of observations
- Predictor variables are highly correlated (collinearity)

Works to decrease variance, which increases bias

- Bias is towards the null
- Shrinkage methods - shrinks the regression coefficients towards zero



LEAST ABSOLUTE SHRINKAGE SELECTION OPERATOR (LASSO)

Same idea as ridge regression

- Both are penalized regression
- Both are shrinkage methods

Use different penalties

- Lasso can zero out coefficients
- Ridge is “easier” to implement, faster to compute



MULTIVARIATE REGRESSION

Multivariate $\neq$ Multi-variable

Multiple response variables in one model

- Model the correlation between response variables
- Same predictors for each, but different relationships
- Don't need special R packages



MULTIVARIATE REGRESSION IN R

```
1 data("airquality")
2 ozone <- airquality[complete.cases(airquality),]
3 m1 <- lm(cbind(Ozone,Solar.R)~Temp+Wind,data=ozone)
4 summary(m1)
```

Response Ozone :

Call:

```
lm(formula = Ozone ~ Temp + Wind, data = ozone)
```

Residuals:

Min	1Q	Median	3Q	Max
-42.156	-13.216	-3.123	10.598	98.492

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-67.3220	23.6210	-2.850	0.00524	**
Temp	1.8276	0.2506	7.294	5.29e-11	***
Wind	-3.2948	0.6711	-4.909	3.26e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1



CLASSIFICATION AND REGRESSION TREES (CART)

Common approach to classification

- Sequential, binary splits of the data
- Classification tree: Categorical data
- Regression tree: Continuous data

Three main elements

- Rules for splitting nodes based on a single variable
- Stopping rules for when no further splits are possible
- Prediction for target variable in each terminal node



ADVANTAGES OF CART

- Non-parametric
- Not significantly impacted by outliers
- Same variables can occur in multiple places
- Can deal with complex inter-dependencies
- Can be used in conjunction with other prediction methods



RANDOM FOREST

Generate bootstrapped trees

- From random sample of the variables
- With replacement

Classify a case for each “tree” in the “forest”

Combine results to decide outcome

- Average for a regression tree
- Majority vote for a classification tree



ALWAYS MORE

Regression is an active field of research

Some models can be re-figured as regression

- Cox proportional hazards model

Plenty I've never done

- Elastic net
- Support vector regression

The tool is out there

